| Pos | Part no. | Description | Qty | Make/buy | Material / spec | kg each |
|---|---|---|---|---|---|---|
| 48 | NM8-800 | Drop cable + base duct (cabling route) | 1 | make | H07RN-F 4G10 + 2 × LiYCY signal; PE duct Ø11 | 12.0 |
| | | F TOWER | | | | |
| 49 | NM8-400 | Tower 8 m, Ø400 → Ø400, t 10/8, one piece | 1 | make | S355J2+N, hot-dip galvanised | 699.8 |
| 50 | NM8-401 | Hand-hole cover 200 × 350 × 6 with gasket | 1 | make | S355 galv. | 3.3 |
| 51 | NM8-410 | Base hinge brackets + hinge pin Ø50 | 1 | make | S355J2+N galv.; pin 42CrMo4+QT | 37.4 |
| | | G FOUNDATION | | | | |
| 52 | NM8-500 | Spread footing (firm ground) with pedestal | 1 | make | C30/37 XF4 air-entrained, B500C | 14880.0 |
| 53 | NM8-510 | Anchor cage 16 × M24 + 4 hinge anchors | 1 | make | M24 8.8 HDG threaded rod, S355 template ring | 161.7 |
| | | H TOOLS AND MOULDS | | | | |
| 54 | NM8-110-1 | Blade mould, lower (pressure side) half on frame | 1 | make | GRP tooling laminate 12 mm, vinylester tooli | 118.1 |
| 55 | NM8-110-2 | Blade mould, upper (suction side) half | 1 | make | GRP tooling laminate 12 mm | 118.1 |
| 56 | NM8-110-3 | Mould frame (RHS 80 × 80 × 4) | 1 | make | S235 RHS 80×80×4 | 178.6 |
| 57 | NM8-420 | Gin pole 3.0 m (erection tool) | 1 | make | S355 tube 168.3 × 6.3 | 94.0 |
| | | J FASTENERS, CABLES, SITE MATERIALS | | | | |
| 58 | NM8-900-01 | Bolt M16 × 90 10.9 zinc-flake + washer (blade roots into the p | 36 | buy | 10.9 zinc-flake | 0.20 |
| 59 | NM8-900-02 | Bolt M16 × 70 10.9 (hub to drum) | 12 | buy | 10.9 zinc-flake | 0.17 |
| 60 | NM8-900-03 | Bolt M16 × 80 10.9 (pitch bearing outer rings to hub necks, 20 | 60 | buy | 10.9 zinc-flake | 0.18 |
| 61 | NM8-900-04 | Bolt M20 × 90 10.9 (spindle to bedplate) | 12 | buy | 10.9 zinc-flake | 0.32 |
| 62 | NM8-900-05 | HV set M16 × 110 10.9 (yaw inner ring, PCD 350) | 16 | buy | EN 14399-4 | 0.26 |
| 63 | NM8-900-06 | Bolt M16 × 70 10.9 (yaw outer ring PCD 490, pads, calipers, br | 40 | buy | 10.9 zinc-flake | 0.17 |
| 64 | NM8-900-07 | HV set M20 × 90 10.9 (adapter to tower) | 16 | buy | EN 14399-4 | 0.40 |
| 65 | NM8-900-08 | Nut M24 8.8 HDG + plate washer (base 16 × 3, hinge 8) | 56 | buy | 8.8 HDG | 0.15 |
| 66 | NM8-900-09 | Small fasteners kit M6–M12 A4 (cover, spinner, mast, nose fram | 1 | buy | A4-70 | 3.00 |
| 67 | NM8-800-W1 | Cable H07RN-F 4G10 (generator → box) | 3 | buy | m | 0.60 |
| 68 | NM8-800-W2 | Cable H07RN-F 4G10 (box → loop → hand hole) | 12 | buy | m | 0.60 |
| 69 | NM8-800-W3 | Cable NYY-J 4 × 10 (hand hole → cabinet) | 4 | buy | m | 0.50 |
| 70 | NM8-800-W4 | Cable ÖLFLEX 4G1.5 shielded (3 brake calipers) | 20 | buy | m | 0.10 |
| 71 | NM8-800-W5 | Cable LiYCY 12 × 0.5 (sensors) | 16 | buy | m | 0.12 |
| 72 | NM8-800-W6 | Cable ÖLFLEX 7G1.5 (yaw motors) | 16 | buy | m | 0.15 |
| 73 | NM8-800-W7 | Slip ring 6 power + 6 signal (pitch supply and bus, in the nos | 1 | buy | | 1.50 |
| 74 | NM8-800-W8 | Earthing conductor Cu 50 mm² + ring earth + 2 rods 3 m | 1 | buy | Cu | 18.00 |
| 75 | NM8-500-M1 | Concrete C30/37 XF4 air-entrained (firm ground) | 6.2 m³ | buy | ready-mix | |
| 76 | NM8-500-M2 | Rebar B500C (mesh top + bottom, pedestal cage) | 1 | buy | ≈ 460 kg | 460.00 |
| 77 | NM8-500-M3 | Non-shrink grout ≥ 40 MPa, 30 mm under the base flange | 70 l | buy | | |
| 78 | NM8-600-K1 | Contactor K1 3-pole NC (mirror contacts) + safety relay PL d | 1 | buy | | 2.00 |
| 79 | NM8-620-PE | Converter 12 kW, dump chopper, AC module, 48 V module, control | 1 | make | in-house electronics | 60.00 |

**NOTES**

1. Masses from the CAD model (cad/turbine_m8/model8.py) for made parts; supplier data or the doc 35 budget for bought and pitch parts. Full BOM: bom8.csv.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

**12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER**

## BILL OF MATERIALS (2/2)

| DRAWING NO. | REV / DATE | SCALE (A3) |
|---|---|---|
| NM8-003 | A · 2026-11-06 | – |
| MATERIAL | MASS (CALC.) | DRAWN / CHECKED |
| see lines | see lines | Claude / not checked |